

One-fresh-root owner quotient and null-cone recession boundary

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1 Purpose and scope firewall

R-115 closed the original stationary scalar $k:2k$ normalizer, but the full one-fresh-root A1 production cluster still contains every visit, every input, complete target heat, mean and trace, mixed predictable baseline, rational recovery, random- W double divergence, and the R-063 contraction forest. The present result pushes that frontier in the complete owner coordinate and identifies the exact finite-dimensional tail condition that any nonlinear root theorem must satisfy.

What is proved. The same-root visit packet descends exactly to one endpoint quotient. The abstract Gaussian partial-Wick decomposition identifies which higher contractions are absent from the physical R-063/R-107 packet. Fresh-root first chaos is absorbed into the frozen whole-output determinant. Actual R-115 conditional packets compose by optimized conditional Holder, and a tilted random- W term is reduced to a doubled-tilt normalizer.

For a finite-dimensional homogeneous quadratic output, the exact Gaussian integrability classifier is a common-null-cone recession number involving the full parabolic tangent image $\mathcal{T}_Q(v)$. Strictly positive recession gives integrability, strictly negative recession gives divergence, and equality genuinely depends on critical-stratum codimension. Two exact centered tensor fixtures show that neither centering nor full Wick ordering alone supplies the required margin.

The exact R-082 pointwise null set is classified. In the finite-Fourier active branch it consists of constant-polarization plane waves, and its real-linearized Pauli–Fierz symbol is explicit. This classification does not prove a quantitative near-null stability estimate or a cutoff-uniform recession margin.

What is not proved. The abstract quadratic theorem is not automatically the rational A1 row. The plane-wave null is a face of a legal production root, not an independently revealable rank-two root atom. No full progressive/revisit production normalizer, one-use source/sextic aggregation, $\text{OVERLAP}_{\text{src}}$, Nelson estimate, removal limit, interacting measure, Sector-A closure, or tier promotion is claimed. A13 remains T4.

The fixed-dimensional algebra is at finite cutoff and fixed positive floor. The exact recession theorem below assumes a polynomial homogeneous quadratic map. The recession of the rational R-082 row is homogeneous of degree two on the sphere but is not globally a quadratic polynomial; applying the same architecture there requires a separately stated homogeneous-map extension and uniform regularity constants.

2 One fresh root: the owner-preserving endpoint quotient

Fix one Gaussian root j , a contraction-closed output cluster C , and a same-root visit path

$$H_0 = A_0, H_1, \dots, H_L = A_*. \quad (2.1)$$

Let

$$x_H = \Pi_C d_j \Xi_H, \quad t_H = P_j \tau_{H,j,C}, \quad (2.2)$$

where the reveal sigma-field, target heat, common covariance, and cluster projection remain unchanged along the path. Put

$$\Delta x_\ell = x_{H_\ell} - x_{H_{\ell-1}}, \quad (2.3)$$

$$\pi_\ell = \Re \langle x_{H_{\ell-1}}, \Delta x_\ell \rangle + \frac{1}{2} \|\Delta x_\ell\|^2 - \frac{1}{2} (t_{H_\ell} - t_{H_{\ell-1}}). \quad (2.4)$$

Theorem 2.1 (same-root visit endpoint quotient)

The complete visit packet telescopes exactly:

$$\boxed{\sum_{\ell=1}^L \pi_\ell = \frac{1}{2} (\|x_{A_*}\|^2 - \|x_{A_0}\|^2) - \frac{1}{2} (t_{A_*} - t_{A_0})}. \quad (2.5)$$

Proof. For every visit,

$$\Re \langle x, \Delta x \rangle + \frac{1}{2} \|\Delta x\|^2 = \frac{1}{2} (\|x + \Delta x\|^2 - \|x\|^2). \quad (2.6)$$

Sum (2.6) and the trace increments. Every internal endpoint occurs once with each sign and cancels. $\square$

For two visits write $a = x_{A_0}$, $f = \Delta x_1$, and $\iota = \Delta x_2$. The square part is

$$\Re \langle a, f \rangle + \frac{1}{2} \|f\|^2 + \Re \langle a + f, \iota \rangle + \frac{1}{2} \|\iota\|^2. \quad (2.7)$$

Thus the fresh-future cross $\Re \langle f, \iota \rangle$ occurs exactly once. Formula (2.7), together with the two trace increments, is precisely the R-107 $a/f/\iota$ packet. This proves quotient descent, not a sign or a normalizer. It also explains why a positive estimate visit by visit is not representation invariant.

3 Partial Wick is a diagnostic, not a replacement owner

Let W be a finite-dimensional isonormal Gaussian process. For scalar polynomials F, G the full Wick product satisfies

$$\boxed{F \diamond G = \sum_{r \geq 0} \frac{(-1)^r}{r!} \langle D^r F, D^r G \rangle}. \quad (3.1)$$

The sum terminates. For Hilbert-valued polynomials the same formula applies componentwise and is basis independent.

Proof. For the exponential vector

$$\mathcal{E}(h) = \exp\{W(h) - \|h\|^2/2\}, \quad (3.2)$$

one has

$$\mathcal{E}(h) \diamond \mathcal{E}(k) = \mathcal{E}(h + k).$$

On the other hand, $D^r \mathcal{E}(h) = h^{\otimes r} \mathcal{E}(h)$, so the right side of (3.1) becomes

$$\mathcal{E}(h) \mathcal{E}(k) e^{-\langle h, k \rangle} = \mathcal{E}(h + k). \quad (3.3)$$

Differentiate in h, k at zero to obtain every polynomial chaos. $\square$

If only root j is Wick ordered and F has root degree N , then for any declared trace owner τ ,

$$\boxed{\frac{1}{2}(\|F\|^2 - \tau) = \frac{1}{2}\langle F \diamond_j F \rangle + \frac{1}{2}(\|D_j F\|^2 - \tau) + \frac{1}{2} \sum_{r=2}^N \frac{(-1)^{r+1}}{r!} \|D_j^r F\|^2.} \quad (3.4)$$

Conditionally on the other roots,

$$\mathbb{E}_j \langle F \diamond_j F \rangle = \|\mathbb{E}_j F\|^2. \quad (3.5)$$

For affine F and $\tau = \|D_j F\|_{\mathfrak{S}_2}^2$, every remainder in (3.4) vanishes. This is the frozen R-107 situation.

Proposition 3.1 (the XY owner firewall)

For independent standard X, Y and $F = XY$,

$$F \diamond F = (X^2 - 1)(Y^2 - 1), \quad (3.6)$$

whereas the production-oriented subtraction of the Y -derivative trace is

$$F^2 - X^2 = X^2(Y^2 - 1). \quad (3.7)$$

Their difference is

$$(F^2 - X^2) - (F \diamond F) = Y^2 - 1. \quad (3.8)$$

This is exactly a covariance times the remaining raw-Wick coordinate, the kind of ΣQ term retained by R-063. Therefore full Wick ordering changes the physical endpoint unless a new production owner theorem proves that all R-063 $P_4/P_2/\Sigma Q/P_0$, P_3/P_1 , trace, and rational companions have recombined to it.

For the R-110 scalar output

$$Y = \varepsilon \{H_2(G) + aH_1(G)\}, \quad (3.9)$$

the physical partial-Wick packet is

$$\frac{1}{2}\{Y^2 - (Y')^2\} = \frac{1}{2}(Y \diamond Y) - \varepsilon^2. \quad (3.10)$$

Full-Wick notation exposes the mean debt $-\varepsilon^2$; it does not pay that debt.

Affine absorption

If the apparently nonlinear fresh-root remainder is actually first chaos, $R(z) = Cz$, it must be merged into the frozen map. With $B = A + C$ and $\delta_0 = \text{Tr}(CC^*)$, the $\rho = 1$ endpoint has the exact form

$$\Phi(1) = -\frac{1}{2} \log \det {}_2(\text{I} + qBB^*) - \frac{q}{2} \langle b, (\text{I} + qBB^*)^{-1} b \rangle. \quad (3.11)$$

Since $x - \log(1 + x) \leq x^2/2$ for $x \geq 0$,

$$\boxed{\Phi(1) \leq \frac{q^2}{4} \|BB^*\|_{\mathfrak{S}_2}^2.} \quad (3.12)$$

Hence all residual first chaos belongs in the frozen whole-output determinant. Only genuine nonlinear chaos needs a new theorem.

4 Composition and the doubled-tilt random- W bridge

Suppose actual conditionally centered packets P_i satisfy, for all $t \geq 0$,

$$\log \mathbb{E}[e^{-tP_i} \mid \mathcal{F}_{i-1}] \leq \frac{t^2 H_i}{4}. \quad (4.1)$$

Conditional Holder with weights $\theta_i > 0$, $\sum_i \theta_i = 1$, gives

$$\log \mathbb{E} e^{-q \sum_i P_i} \leq \frac{q^2}{4} \sum_i \frac{H_i}{\theta_i}. \quad (4.2)$$

The optimizer is $\theta_i = \sqrt{H_i} / \sum_k \sqrt{H_k}$, hence

$$\log \mathbb{E} e^{-q \sum_i P_i} \leq \frac{q^2}{4} \left(\sum_i \sqrt{H_i} \right)^2. \quad (4.3)$$

At $q = 10/9$, $q^2/4 = 25/81$. Conditional independence improves the right side to $(25/81) \sum_i H_i$. Identical same-root copies show that this additive improvement cannot follow from marginal R-115 estimates alone.

Let ℓ be the complete nonlinear packet and let W be the same-root random covariance from R-110. With

$$d\nu = \frac{e^{-q\ell}}{\mathbb{E} e^{-q\ell}} d\gamma,$$

Gaussian integration by parts gives

$$q^2 \mathbb{E}_\nu \langle D\ell, W D\ell \rangle - q \mathbb{E}_\nu \text{Tr}(W D^2 \ell) = \mathbb{E}_\nu \delta^2 W. \quad (4.4)$$

Writing $K_W = \mathbb{E}(\delta^2 W)^2$, Cauchy–Schwarz under the original Gaussian law yields

$$|\mathbb{E}_\nu \delta^2 W| \leq \sqrt{K_W} \exp \left\{ \frac{\Phi(2q) - 2\Phi(q)}{2} \right\}. \quad (4.5)$$

If $\Phi(t) \leq t^2 H_*/4$, then

$$|\mathbb{E}_\nu \delta^2 W| \leq \sqrt{K_W} e^{q^2 H_*/2}. \quad (4.6)$$

The exponent is $50H_*/81$ at $q = 10/9$. Thus the random- W bridge requires the same doubled-tilt normalizer that the root theorem is meant to provide; K_W alone is not a tilted bound.

The interpolation endpoint and its cross-score also cannot be budgeted as independent positive objects. Already for the affine scalar fixture with $A = C = 1$ and trace path $t_\rho = 2 + 2\rho$,

$$\Phi_\rho = \frac{qt_\rho - \log(1 + qt_\rho)}{2} - \frac{qb^2}{2(1 + qt_\rho)}. \quad (4.7)$$

The increment from $\rho = 0$ to 1 contains a positive multiple of b^2 , although the tangent covariance costs do not depend on b . The negative resolvent term in the total endpoint is the cancellation. Separating it from the interpolation cross-score destroys that cancellation.

5 Exact quadratic recession classifier

Let $X \sim N(0, I_n)$, let $Q : \mathbb{R}^n \rightarrow H$ be homogeneous quadratic, let $A : \mathbb{R}^n \rightarrow H$ be linear, let $b \in H$, and let τ be a continuous scalar homogeneous quadratic form. Define

$$L(x) = \frac{1}{2} \{ \|b + Ax + Q(x)\|^2 - \tau(x) \}, \quad (5.1)$$

$$U_q(x) = \frac{1}{2}\{|x|^2 - q\tau(x) + q\|b + Ax + Q(x)\|^2\}. \quad (5.2)$$

Then $\mathbb{E}e^{-qL(X)}$ is a constant times $\int e^{-U_q(x)}dx$.

Put $Z_Q = Q^{-1}(0)$. For $v \in Z_Q \cap S^{n-1}$ define the parabolic tangent image

$$\mathcal{T}_Q(v) = \{z : \exists r_j \rightarrow \infty, v_j \rightarrow v, |v_j| = 1, r_j Q(v_j) \rightarrow z\}. \quad (5.3)$$

Theorem 5.1 (exact parabolic-tangent recession)

With $\inf \emptyset = +\infty$,

$$\lambda_q := \liminf_{|x| \rightarrow \infty} \frac{2U_q(x)}{|x|^2} = \inf_{\substack{v \in Z_Q \cap S^{n-1} \\ z \in \mathcal{T}_Q(v)}} \left\{ 1 - q\tau(v) + q\|Av + z\|^2 \right\}. \quad (5.4)$$

Moreover:

$$\lambda_q > 0 \implies \mathbb{E}e^{-qL(X)} < \infty, \quad (5.5)$$

$$\lambda_q < 0 \implies \mathbb{E}e^{-qL(X)} = \infty. \quad (5.6)$$

The case $\lambda_q = 0$ is not decided by quadratic recession.

Proof. For $x = rv$, $|v| = 1$,

$$\frac{2U_q(rv)}{r^2} = 1 - q\tau(v) + q\left\|Av + rQ(v) + \frac{b}{r}\right\|^2. \quad (5.7)$$

Along a finite recession sequence, compactness supplies a convergent subsequence $v_j \rightarrow v$. Finiteness forces $Q(v_j) \rightarrow 0$ and boundedness of $r_j Q(v_j)$, producing (5.3). Conversely, every sequence in (5.3) inserted in (5.7) realizes its displayed recession value. This proves (5.4). If $Z_Q \cap S^{n-1}$ is empty, compactness gives $\inf_S \|Q\| > 0$, and both sides are $+\infty$.

If $\lambda_q > 0$, then outside a ball $U_q(x) \geq \lambda_q|x|^2/4$, proving (5.5). If $\lambda_q < 0$, select x_j with $U_q(x_j) \leq -c|x_j|^2$. Since U_q is a polynomial of degree at most four, $|\nabla U_q| \leq C|x|^3$ on comparable-radius annuli. A ball of radius $\delta/|x_j|$ therefore retains $U_q \leq -c|x_j|^2/2$. Pass to a disjoint subsequence. Its integral is bounded below by

$$\sum_j C|x_j|^{-n}e^{c|x_j|^2/2} = \infty, \quad (5.8)$$

which proves (5.6). $\square$

The predictable baseline b disappears only from strict quadratic recession. It can decide the equality case through lower-order terms.

Why the derivative range is not enough

Always

$$\text{Ran } DQ(v) \subseteq \mathcal{T}_Q(v). \quad (5.9)$$

At a singular null, if $DQ(v)\zeta = 0$, then second-order approaches can also produce

$$DQ(v)\eta + Q(\zeta) \in \mathcal{T}_Q(v). \quad (5.10)$$

For example,

$$Q(x_1, x_2, x_3) = (x_1x_2, x_3^2), \quad v = e_1, \quad (5.11)$$

has $\text{Ran } DQ(v) = \text{span}\{(1, 0)\}$, while

$$v_r = (\sqrt{1 - r^{-1}}, 0, r^{-1/2})$$

gives $rQ(v_r) \rightarrow (0, 1)$. Thus replacing $\mathcal{T}_Q(v)$ by the derivative range is false without an error bound.

A sufficient local hypothesis is the metric regularity estimate

$$\text{dist}(w, Z_Q \cap S^{n-1}) \leq C\|Q(w)\| \quad (5.12)$$

near v , together with continuity of DQ . Then

$$\mathcal{T}_Q(v) = \text{Ran } DQ(v), \quad (5.13)$$

and the inner infimum becomes

$$\text{dist}\{-Av, \text{Ran } DQ(v)\}^2 = \text{dist}\{Av, \text{Ran } DQ(v)\}^2. \quad (5.14)$$

Without linearity of the tangent image, the safe expression is $\text{dist}(-Av, \mathcal{T}_Q(v))^2$; its sign cannot be silently changed.

An unconditional but stronger sufficient condition is

$$\sup_{v \in Z_Q \cap S^{n-1}} q\tau(v) < 1. \quad (5.15)$$

Proposition 5.2 (equality depends on codimension)

Let $x = (t, y) \in \mathbb{R} \times \mathbb{R}^k$ and

$$Q(t, y) = ty, \quad A = b = 0, \quad \tau(t, y) = t^2/q. \quad (5.16)$$

Then $\lambda_q = 0$ and, up to a finite dimensional constant,

$$\mathbb{E}e^{-qL} = \int_{\mathbb{R}} (1 + qt^2)^{-k/2} dt. \quad (5.17)$$

This diverges for $k = 1$ and is finite for $k > 1$. Thus a zero recession margin requires the critical-stratum dimension, normal rank, lower-order terms, and singular strata; a sign check alone cannot close the endpoint.

6 Two centered tensor no-gos

The following fixtures are abstract Gaussian output tensors. They are not claimed to be legal A1 rows. Their role is to eliminate universal routes before production ownership is distorted to fit them.

Proposition 6.1 (exactly centered quadratic null-cone divergence)

Let U, V be independent standard Gaussians and set

$$Q(U, V) = U^2 - V^2, \quad b = 2, \quad \tau(U, V) = 4(U^2 + V^2), \quad (6.1)$$

$$L = \frac{1}{2}\{(2 + Q)^2 - \tau\}. \quad (6.2)$$

Then

$$\mathbb{E}L = 0. \quad (6.3)$$

With $W = DQ \otimes DQ$,

$$W = 4 \begin{pmatrix} U^2 & -UV \\ -UV & V^2 \end{pmatrix}, \quad (6.4)$$

$$\mathbb{E}\|W\|_{\mathfrak{S}_2}^2 = 128, \quad (6.5)$$

$$\delta^2 W = 4(U^2 - V^2)^2 - 12(U^2 + V^2) + 8, \quad (6.6)$$

$$\mathbb{E}(\delta^2 W)^2 = 1088. \quad (6.7)$$

Nevertheless

$$\boxed{\mathbb{E}e^{-qL} = \infty \quad \text{for every } q \geq \frac{1}{4}.} \quad (6.8)$$

Proof. Put $u = (U + V)/\sqrt{2}$ and $v = (U - V)/\sqrt{2}$. Then

$$L = 2 + 4uv + 2u^2v^2 - 2u^2 - 2v^2. \quad (6.9)$$

On $|u| \leq |v|^{-1}$,

$$L \leq 8 - 2v^2. \quad (6.10)$$

After including the Gaussian density, the lower exponential rate is $(2q - 1/2)v^2$. It is positive for $q > 1/4$. At equality the tube width is proportional to $1/|v|$, giving logarithmic divergence. Exact Gaussian moments give (6.3), (6.5), and (6.7). $\square$

Thus exact centering, positive semidefinite W , and finite unweighted H and K_W do not imply the target normalizer. A strict common-null-cone margin or an equality-stratum theorem is indispensable.

Proposition 6.2 (full Wick ordering still has a sharp boundary)

Let

$$R(U, V) = (2\varepsilon UV, 2\kappa\varepsilon H_2(V)). \quad (6.11)$$

Its abstract full-Wick packet is

$$\mathcal{P}_\kappa = 2\varepsilon^2 \{H_2(U)H_2(V) + \kappa^2 H_4(V)\}, \quad (6.12)$$

so $\mathbb{E}\mathcal{P}_\kappa = 0$, and

$$\mathbb{E}\|DR, DR^*\|_{\mathfrak{S}_2}^2 = 128\varepsilon^4(1 + \kappa^2 + 6\kappa^4). \quad (6.13)$$

Writing $\alpha = q\varepsilon^2$, conditioning on V gives

$$M_\kappa(\alpha) = \mathbb{E}_V \frac{\exp\{2\alpha(V^2 - 1) - 2\alpha\kappa^2 H_4(V)\}}{\sqrt{1 + 4\alpha(V^2 - 1)}}. \quad (6.14)$$

Its exact domain is

$$\boxed{0 \leq \alpha < \frac{1}{4}.} \quad (6.15)$$

At equality the integrand is proportional to $1/|V|$ near zero; above it the conditional U integral diverges on a positive-measure interval.

For a completely explicit $q = 10/9$ violation take

$$\kappa = \frac{1}{100}, \quad \delta = 10^{-50}, \quad \alpha = \frac{1 - \delta}{4}, \quad \varepsilon^2 = \frac{9(1 - \delta)}{40}. \quad (6.16)$$

The proposed covariance target is

$$\frac{q^2}{4} \mathbb{E}\|DR, DR^*\|_{\mathfrak{S}_2}^2 = 32\alpha^2(1 + \kappa^2 + 6\kappa^4) < \frac{21}{10}. \quad (6.17)$$

On $\sqrt{\delta} \leq |V| \leq 1$, the denominator in (6.14) is at most $\sqrt{2}|V|$ and the exponent including the Gaussian density is greater than -1 . Hence

$$M_\kappa(\alpha) > \frac{25 \log 10}{e\sqrt{\pi}} > \frac{115}{12} > 9 > e^{21/10}. \quad (6.18)$$

This disproves the universal full-Wick covariance target despite exact centering. It does not embed an A1 Cartan/Fierz tensor, heat trace, Fourier owner, chronology, rational recovery, or R-063 forest.

7 The exact R-082 null set and the production scope correction

Write the doublet as $u = (u_1, u_2) \in \mathbb{C}^2$, the singlet as $\chi \in \mathbb{C}$, and a tangent as (v, w) . Put

$$r = |u|^2, \quad \rho = r + |\chi|^2, \quad d = \rho + e, \quad (7.1)$$

$$dr = 2\Re(u^*v), \quad d\rho = dr + 2\Re(\bar{\chi}w), \quad h = u_1v_2 - u_2v_1. \quad (7.2)$$

With $c_0, c_1 > 0$ and $\alpha = 5/9$, the R-082 current coordinate is

$$\Xi(u, \chi; v, w) = \begin{pmatrix} \sqrt{c_0}, dr \\ \sqrt{c_1}\{dr - \alpha(r/d)d\rho\} \\ 2\sqrt{c_0 + c_1}\Re h \\ 2\sqrt{c_0 + c_1}\Im h \end{pmatrix}. \quad (7.3)$$

Theorem 7.1 (pointwise and finite-Fourier null classification)

At one point,

$$\Xi = 0 \iff dr = 0, \quad h = 0, \quad \frac{r}{d}d\rho = 0. \quad (7.4)$$

Consequently:

1. if $u = 0$, every tangent (v, w) is null;
2. if $u \neq 0$, then $v = i\theta u$ for real θ and $\Re(\bar{\chi}w) = 0$; when $\chi \neq 0$, $w = i\eta\chi$ for real η .

Let a smooth field on a connected torus satisfy $\Xi(\Psi, \partial_i\Psi) = 0$ in every spatial direction. Either $u \equiv 0$, in which case χ is unrestricted, or

$$u(x) = pf(x), \quad |f(x)| = \text{constant}, \quad |\chi(x)| = \text{constant}. \quad (7.5)$$

If f and χ have exact finite Laurent support, then the active branch is

$$\boxed{u(x) = pe^{ik \cdot x}, \quad \chi(x) = ce^{i\ell \cdot x}}, \quad (7.6)$$

with independent lattice wavevectors and with $c = 0$ allowed.

Proof. Positivity of the four weights proves (7.4). If $u \neq 0$, $h = 0$ makes v complex-collinear with u , while $dr = 0$ makes its coefficient purely imaginary. Along a field, $dr_i = 0$ makes $|u|$ constant. On the active branch $h_i = 0$ fixes the projective direction, and the rational row fixes $|\chi|$. For a finite Laurent polynomial f , constant modulus on the torus gives

$$f(z)f^\#(z) = c^2 \quad (7.7)$$

as a Laurent-polynomial identity. Thus f/c is a unit of $\mathbb{C}[z_1^{\pm 1}, \dots, z_d^{\pm 1}]$. Laurent units are monomials: equivalently, the Newton polytope of a product is the Minkowski sum, and the sum in (7.7) is a point only when each polytope is a point. Apply the same argument to a nonzero χ . $\square$

This theorem uses every spatial derivative direction, the full unprojected current, a connected torus, exact finite Laurent support, and non-aliased products. It is false as stated for one direction, a projected output cluster, or the pure-singlet branch.

The real-linearized null symbol

At (7.6), for a perturbation (a, s) define

$$R_i = 2\Re\{u^*(\partial_i - ik_i)a\}, \quad (7.8)$$

$$T_i = R_i + 2\Re\{\bar{\chi}(\partial_i - i\ell_i)s\}, \quad (7.9)$$

$$H_i = u_1(\partial_i - ik_i)a_2 - u_2(\partial_i - ik_i)a_1. \quad (7.10)$$

Then

$$D\Xi_i(a, s) = \begin{pmatrix} \sqrt{c_0}R_i \\ \sqrt{c_1}(R_i - \alpha\vartheta T_i) \\ 2\sqrt{c_0 + c_1}\Re H_i \\ 2\sqrt{c_0 + c_1}\Im H_i \end{pmatrix}, \quad (7.11)$$

where $\vartheta = r/(\rho + e)$ for the exact floor and $\vartheta = r/\rho$ for the homogeneous recession row.

This symbol is real-linear: the real parts couple Fourier modes m and $-m$. Its kernel is larger than the exact finite-Laurent null manifold. For real phase modulations $a = i\varphi u$, $s = i\psi\chi$, all rows in (7.11) vanish. Such phase-modulated fields need not remain in the same finite Fourier space. Therefore Laurent rigidity does not imply first-order coercivity and does not establish $\mathcal{T}_Q(v) = \text{Ran } DQ(v)$. A quantitative metric error bound must be proved separately, uniformly in the cutoff.

The gauge face is not a standalone production root

The deterministic field

$$u_1 = x_1 \cos(kx) + ix_2 \sin(kx), \quad u_2 = \chi = 0 \quad (7.12)$$

has

$$dr = k(x_2^2 - x_1^2) \sin(2kx), \quad d\rho = dr, \quad h = 0. \quad (7.13)$$

On $x_1 = \pm x_2 = a$ it is the exact plane wave $u_1 = ae^{\pm ikx}$ and all four rows vanish. The positive floor does not lift this phase-gauge kernel.

However, retaining only the real-cosine and imaginary-sine coordinates violates the componentwise common-even production parity

$$\mathbb{E}[X^c \partial_i X^a] = 0. \quad (7.14)$$

The omitted equal-variance sine/cosine partners restore (7.14). Hence the rank-two face is not an admissible independently revealed R-104 root. Every complementary polarization, internal component, shared output, future feedback, rational harmonic, and R-063 contraction from the same root must remain in one conditional cluster.

Let $D_{aa,j,i}$ be the complex derivative covariance of the full legal root. In the six-real convention $\Gamma_{\mathbb{R}} = \frac{1}{2} \text{realify}(D_{\mathbb{C}})$. On the null field, the full-root trace is

$$\tau_{j,\text{null}}(a) = a^2 \sum_i [2A(a)D_{11,j,i} + 4(c_0 + c_1)D_{22,j,i}], \quad (7.15)$$

$$A(a) = c_0 + c_1 \left(1 - \frac{5}{9} \frac{a^2}{a^2 + e}\right)^2. \quad (7.16)$$

The packet is $P_{j,\text{null}} = -\tau_{j,\text{null}}/2$. This refutes pointwise current-square coercivity, but not the full-root Gaussian normalizer.

Indeed, for the uncontrolled full stationary production Gaussian, componentwise value–gradient parity and joint Gaussianity make the value and gradient independent. With the rational row included,

$$\mathbb{E} \left[\frac{1}{2} |\Xi(U, \partial U)|^2 - \frac{1}{2} B(U) : \Gamma \right] = 0. \quad (7.17)$$

The complementary root coordinates restore centering in expectation.

Since

$$c_0 + c_1 = \frac{339}{8000P}, \quad (7.18)$$

define

$$K_j^{\max} = 2(c_0 + c_1) \sum_i D_{11,j,i} + 4(c_0 + c_1) \sum_i D_{22,j,i}. \quad (7.19)$$

At fixed cutoff,

$$P_{j,\text{null}}(a) + \frac{3}{20}a^6 \geq \frac{3}{20}a^6 - \frac{1}{2}K_j^{\max}a^2 \geq -\frac{3}{10} \left(\frac{10}{9}K_j^{\max} \right)^{3/2}. \quad (7.20)$$

The derivative point covariance grows with cutoff, so this loss is not uniform; it may scale as the cutoff radius to the 3/2 power. Spending the terminal sextic once for every root would also duplicate its owner.

8 The exact remaining certificate

For a fixed standardized polynomial quadratic root packet, Theorem 5.1 says that the strict finite-dimensional target is

$$\inf_{\substack{v \in Z_{Q_N}, \ \|v\|_\gamma=1 \\ z \in \mathcal{T}_{Q_N}(v)}} \left[1 - \frac{10}{9}\tau_N(v) + \frac{10}{9}\|A_N v + z\|_{\text{PF}}^2 \right] > 0. \quad (8.1)$$

Sector A needs far more than positivity at each cutoff. It needs one predictable margin uniform in cutoff and floor, together with uniform derivative and error-bound constants. It must cover:

1. the pure-singlet null subspace and its degeneration from the active branch;
2. every active plane-wave parameter p, c, k, ℓ and the realified $m/-m$ coupling;
3. singular parabolic tangent images, not merely $\text{Ran } DQ_N(v)$;
4. covariance standardization and the complete heat-lifted trace;
5. mixed predictable baselines, every visit and input, rational recovery, random- W double divergence, and the R-063 forest exactly once;
6. a single global use of source energy and terminal sextic.

For the rational-homogeneous R-082 recession map, (8.1) is a target schema, not yet an application of the polynomial theorem. A valid proof may instead establish the stronger trace margin

$$\sup_N \sup_{Q_N(v)=0, \ \|v\|_\gamma=1} \tau_N(v) < \frac{9}{10}, \quad (8.2)$$

or extend Theorem 5.1 to the smooth homogeneous rational map and prove its full parabolic-tangent margin. Equality in either route still requires a critical-stratum normal-rank analysis.

This isolates the remaining A13 work into two load-bearing gates:

1. the complete contraction-closed progressive/revisit production cluster with all endpoint owners retained;
2. the global one-use source/sextic aggregation after that local cluster is controlled.

Only after those gates can $\text{OVERLAP}_{\text{src}}$, Nelson, removals, and the interacting measure be resumed.

9 Proof-route evidence map

Route	Verdict	Research-reusable reason
same-root positive sub-visits	superseded	only the recombined endpoint (2.5) descends to the representation quotient
full Wick replacement	failed	it changes the R-063 owner; (3.8) is a missing ΣQ term
absorb first chaos	accepted	merge it into $B = A + C$ and use the exact whole-output determinant
marginal packet multiplication	boundary	Holder gives square-of-sum cost; additive cost needs conditional independence
unweighted random- W cost	boundary	the tilted term requires the doubled normalizer in (4.5)
centering alone	failed	Proposition 6.1 has exact mean zero and divergent normalizer
full-Wick centering	failed	Proposition 6.2 reaches the sharp $\alpha = 1/4$ null boundary
ray-only tail test	superseded	the exact classifier uses the full parabolic tangent image (5.4)
Laurent null rigidity	accepted with boundary	exact nulls are classified, but near-null metric stability is still open
rank-two gauge root	withdrawn	it is a deterministic face, not a temporally faithful independently revealable production root
rootwise sextic repair	finite-cutoff only	(7.20) is not cutoff uniform and duplicates the terminal owner if repeated

The executable evidence checks the endpoint identities, Wick owner defect, affine determinant, Holder and doubled-tilt constants, both centered tensor moments and thresholds, critical-codimension integrals, R-082 plane-wave null, real-linearized symbol, and finite-cutoff sextic minimum. A structurally independent route reconstructs them without importing the primary program.

10 Devil’s-advocate review

1. **Objection: “The endpoint quotient proves a normalizer.” VALID WITH MITIGATION.** Equation (2.5) is an exact owner identity only. No sign or exponential estimate follows from it.
2. **Objection: “Full Wick ordering repairs the physical packet.” UPHELD AGAINST THAT ROUTE.** Equations (3.6)–(3.8) exhibit the missing ΣQ owner, and Proposition 6.2 shows that even the changed abstract endpoint has a sharp null boundary.
3. **Objection: “Exact centering and finite H, K_W are enough.” UPHELD AGAINST THAT ROUTE.** Proposition 6.1 has mean zero, $H = 128$, $K_W = 1088$, and divergence already at $q = 1/4$.
4. **Objection: “The ray value $\|Av\|$ is the tail criterion.” UPHELD AGAINST THAT ROUTE.** Singular second-order approaches enlarge $\mathcal{T}_Q(v)$, as (5.11) shows. The exact sign is (5.4).
5. **Objection: “The derivative-range distance formula is always equivalent.” UPHELD AGAINST THAT ROUTE.** It requires the metric error bound (5.12). Without a linear tangent image the safe distance is to $-Av$, not to Av .

6. **Objection:** “The finite-Laurent null classification supplies coercivity.” **UPHELD AGAINST THAT ROUTE.** Real phase modulations lie in the linearized kernel, and near-null parabolic sequences are not controlled by exact-zero rigidity.
7. **Objection:** “The plane-wave face is a legal rank-two root counterexample.” **UPHELD AGAINST THE EARLIER WORDING.** Componentwise parity fails after that coordinate deletion. The face proves pointwise noncoercivity only; the full stationary root is centered.
8. **Objection:** “The six-real trace lost a factor two.” **DISMISSED.** Formula (7.15) uses $\Gamma_{\mathbb{R}} = \text{realify}(D_{\mathbb{C}})/2$, includes the complementary u_1 polarization, and retains the u_2 determinant derivative even at realized $u_2 = 0$.
9. **Objection:** “The terminal sextic repairs every root.” **UPHELD AGAINST THAT ROUTE.** The exact fixed-cutoff minimum is (7.20), but $K_j^{\max}$ is not cutoff uniform and the terminal sextic is a one-use global owner.
10. **Objection:** “The polynomial recession theorem already covers the A1 rational row.” **UPHELD AGAINST THAT WORDING.** The rational row is only degree-two homogeneous at recession. A smooth homogeneous extension with uniform constants remains to be proved.

11 Result footer

Result ID: A13-CLASSII-ONE-FRESH-ROOT-OWNER-QUOTIENT-WICK-NULLCONE-BOUNDARY

Ledger: R-116.

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.

Precise statement: Theorems 2.1, 5.1, and 7.1; Propositions 3.1, 5.2, 6.1, and 6.2; affine absorption (3.11)--(3.12); conditional composition (4.1)--(4.6); production scope correction (7.12)--(7.20).

Scope: finite-dimensional polynomial quadratic recession theorem; finite cutoff and fixed floor for A1 null algebra; abstract tensor no-gos are not production embeddings.

Dependencies: R-063, R-072, R-079, R-082, R-105, R-107, R-110, R-115.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command:

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/
a13_classii_one_fresh_root_owner_quotient_wick_nullcone_boundary_verify.py
```

Expected output: Integrated R-116 PASS; primary 30/30 and independent 23/23 children PASS, with all source, note, PDF, and manifest pins valid.

Falsification gate: any failed endpoint, Wick, recession, tensor-moment, critical-codimension, production-symbol, trace-factor, child/source/note, PDF, or manifest assertion; any use beyond the stated scope.

Tier before / after: T4 / T4.

Open gates: A13-CLASSII-FULL-PROGRESSIVE-REVISIT-EXTENSION;

A13-CLASSII-CONTROLLED-SHELL-ENERGY-ONE-USE.

No-overclaim statement: No full A1 conditional normalizer, one-use source/sextic aggregation, OVERLAP_src, Nelson estimate, removal limit, interacting measure, Sector-A closure, or tier promotion is established.

Next required action: Prove a cutoff-uniform parabolic-tangent or stronger trace margin for the complete rational-homogeneous production root while retaining all owners, then aggregate source and terminal sextic once.